Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Integrated AI-Driven Orchestration Platform for Self-Sustaining Battery Swapping Networks: Synergizing MARL-based Indirect Rebalancing and Demand-Responsive Incremental Siting			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Energy & Resources	
	Medium Category	Small Category	Medium Category	Small Category
	Manufacturing Value-Added Services	Distribution and Logistics Services	*Smart Grid*	Intelligent Service - Operation
Concept of Proposed Technology	The proposed technology aims to develop a **unified AI-driven operating platform for Electric Two-Wheeler (E2W) Battery Swapping Services (BaaS)**, specifically optimized for the high-frequency delivery logistics sector. The system achieves a "logistics-free" self-rebalancing network by leveraging advanced Reinforcement Learning (RL) and Multi-Agent Reinforcement Learning (MARL) to optimize charging and nudge user behavior.			

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	Key features: • **Opportunity Cost-Aware MARL Incentives:** A Multi-Agent Reinforcement Learning engine that calculates real-time incentives based on a rider's delivery route and time-based earnings potential to encourage voluntary battery relocation. • **Demand-Responsive Smart Charging:** An RL-based scheduler that synchronizes with food delivery peak hours (lunch/dinner) to ensure 99% battery availability. • **AI-Powered Incremental Siting:** A multi-period facility location model that recommends optimal expansion sites for new stations based on delivery order heatmaps and rider movement trajectories.
Rationale for Joint R&D Project	This project leverages the unique synergy between the world's most dense delivery market (Korea) and top-tier logistics optimization expertise (Georgia Tech). Rationale for collaboration: • **The Research Institute (Georgia Tech)** provides world-class expertise in stochastic optimization, MARL for behavioral modeling of decentralized agents (riders), and large-scale SCM algorithms. • **The Korean SME Partner** provides the essential BaaS platform infrastructure, real-world operational data, hardware-software integration capabilities, and the commercial platform required for field validation. Expected benefits of joint R&D: • This collaboration will accelerate the deployment of cost-effective, self-sustaining battery networks, reducing the high capital and operational expenditures.
Key Words	1. Gig Economy 2. Battery Swapping Service (BaaS) 3. Multi-Agent Reinforcement Learning (MARL) 4. Indirect Rebalancing 5. Incremental Siting

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The primary objective is to develop an autonomous operating engine that maintains system equilibrium through incentivized rider behavior and maximizes service availability through data-driven expansion and user-led battery inventory management. The specific R&D objectives are: ● To develop a **Rider-Behavioral RL Engine** that minimizes detour distance ($<500m$) during active delivery missions. ● To implement a **Predictive Inventory Management System** that forecasts battery depletion using delivery order data. ● To formulate an **Incremental Expansion Strategy** that identifies high-ROI nodes in an evolving urban network.			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Swap Success Rate (SSR)** Achieve > 98.5% availability during peak hours. **Indirect Rebalancing rate** Resolve > 90% of station imbalances via user nudging **Operating Expense Reduction** Achieve > 15% reduction in total operating costs (energy + logistics) compared to traditional models.			

R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from simulation to laboratory environments and finally to real-world domestic settings.
	
	● Year 1: Simulation-Based Core AI Development
	  ○ Implement gig-worker-centric demand forecasting and initial RL/MARL models for on-route indirect rebalancing within a high-fidelity simulation environment.
	  ○ Develop a multi-period incremental siting algorithm for initial network expansion by analyzing food delivery order heatmaps and historical rider trajectory data.
	● Year 2: Laboratory Integration and Multi-Agent Optimization
	  ○ Integrate charging, pricing, and siting modules into a unified decision-support engine capable of balancing rider opportunity costs with system-wide inventory requirements.
	  ○ Conduct high-fidelity simulations to evaluate the stability and convergence of indirect rebalancing under various behavioral models of price-sensitive delivery riders.
	● Year 3: Real-World Testbed Deployment and Validation
	  ○ Deploy the integrated platform in a high-density Korean urban testbed provided by the SME partner to validate the "logistics-free" operational model in an active delivery environment.			

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	○ Refine algorithms based on actual field data to maximize the efficiency of crowdsourced rebalancing and finalize the commercialization model for the global Battery-as-a-Service (BaaS) market.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute (Georgia Tech) will be responsible for the theoretical design, algorithmic development, and simulation-based validation of the core AI and optimization engines. Georgia Tech will play a pivotal role in establishing the mathematical foundations for decentralized battery management and user-led inventory balancing. The specific roles and responsibilities of Georgia Tech include: ● **Development of Multi-Agent Reinforcement Learning (MARL) algorithms** for dynamic pricing and incentivized user-led indirect rebalancing. ● **Design of predictive demand models** that integrate third-party delivery order data with battery consumption patterns. ● **Design and formulation of stochastic optimization models** for multi-period incremental station siting and network capacity expansion. ● **Provision of technical expertise on algorithm scalability and convergence** for large-scale urban infrastructure networks comprising thousands of nodes.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the battery swapping station (BSS) hardware and system-level implementation required for the proposed self-sustaining operating platform. The SME will play a key role in translating advanced AI algorithms into a practical and deployable urban mobility service. The specific roles and responsibilities of the Korean SME partner include: ● **System integration of AI-driven charging and pricing algorithms** with BSS hardware, cloud-based management servers, and embedded software platforms. ● **Development of rider-facing mobile application interfaces** to deliver real-time navigation nudges and reward notifications during delivery. ● **Selection and management of urban testbeds** in Korea, including the collection of high-fidelity GPS and battery usage data from delivery fleets. ● **Refinement of technology for commercial scalability**, ensuring hardware reliability and regulatory compliance for the domestic and global markets.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project is expected to deliver an **AI-driven orchestration platform specifically optimized for the delivery-heavy urban mobility landscape**. The project will advance the state of the art in **gig-worker behavioral modeling and logistics-free battery inventory rebalancing**, achieving measurable performance improvements in service availability (SSR) and operational efficiency. From an industrial perspective, this project will provide the Korean SME partner with **core AI-enabled Battery-as-a-Service (BaaS) technologies** and system integration know-how applicable to next-generation electric mobility infrastructure. The outcomes will support technology commercialization and product development to enable **cost-effective and scalable battery-swapping solutions**. More broadly, the project is expected to **strengthen the competitiveness of Korean battery and mobility SMEs** and contribute to the growth of the sustainable energy ecosystem through international collaboration. By establishing a "logistics-free" operational benchmark, the partnership will position the participating SME as a leader in the global smart city and advanced mobility markets, fostering innovation in zero-emission urban transportation.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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